Gavin's Lectures on Physics - Electrodynamics



Author：长颈鹿 Gavin

October 19, 2026

Contents

1 Maxwell Equations **3**
- 1.1 Maxwell Equations . . . 3
- 1.2 d'Alembert Equation . . . 3
- 1.3 Covariant Maxwell Equations . . . 4

2 Dielectric and Magnetic Media **5**
- 2.1 Electric Polarization . . . 5
- 2.2 Magnetization . . . 5

3 Action Principle **7**
- 3.1 Action of a Charged Particle in EM Field . . . 7
- 3.2 Action of the Electromagnetic Field . . . 7

4 Lorentz Transformation of EM Field **9**

5 Static Fields **11**
- 5.1 Static Electric Field . . . 11
- 5.2 Static Magnetic Field . . . 11
- 5.3 Ferromagnetism (铁磁性) . . . 12
  - 5.3.1 Magnetic Media (磁介质) . . . 12
  - 5.3.2 Ferromagnetism (磁畴) . . . 12
  - 5.3.3 Hysteresis Loop (磁滞回线) . . . 12
  - 5.3.4 Magnetic Circuit . . . 13

6 Electromagnetic Wave **14**
- 6.1 Wave Equation and Basic Properties . . . 14
- 6.2 Telegrapher Equation . . . 14
- 6.3 Energy and Energy Flux (Poynting Theorem) . . . 15
- 6.4 Interface Phenomena: Refraction, Reflection . . . 15
- 6.5 Interference and Diffraction . . . 16
- 6.6 Polarization . . . 17

7 Electromagnetic Radiation **18**
- 7.1 Point Charge Radiation . . . 18
  - 7.1.1 Liénard–Wiechert Potentials . . . 18
  - 7.1.2 Non-Relativistic Radiation Fields . . . 19
  - 7.1.3 Relativistic Radiation Fields: Poynting Vector and Power . . . 22
  - 7.1.4 Complete Liénard–Wiechert Fields . . . 22
- 7.2 Electric Dipole Radiation . . . 23

8 Radiation in Matter **25**
- 8.1 Wave in Plasma: Free Electron . . . 26
  - 8.1.1 Dispersion relation of plasma . . . 27
  - 8.1.2 Free-electron scattering . . . 27
- 8.2 Wave in Dielectrics（电介质）: Bound Electron . . . 28
  - 8.2.1 Dispersion relation of dielectrics . . . 29
  - 8.2.2 Bound-electron scattering . . . 29
- 8.3 Wave in Conductors . . . 30

9 Kramers–Kronig Relation **32**
9.1 Cauchy Integral Formula 32
9.2 Sokhotski–Plemelj Equation 33
9.3 Kramers–Kronig Relation 33

1 Maxwell Equations

Maxwell equations can be expressed in 3 equivalent formulations.

1.1 Maxwell Equations

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t} \end{cases} \tag{1}$$

In linear, homogeneous, isotropic media

$$\begin{cases} \mathbf{J} = \sigma \mathbf{E} \\ \mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 (1 + \chi_e) \mathbf{E} \\ \mathbf{B} = \mu \mathbf{H} = \mu_0 \mu_r \mathbf{H} = \mu_0 (\mathbf{H} + \mathbf{M}) = \mu_0 (1 + \chi_m) \mathbf{H} \end{cases} \tag{2}$$

Boundary Conditions

$$\begin{cases} \hat{n} \cdot (\mathbf{D_2} - \mathbf{D_1}) = \sigma_f \\ \hat{n} \cdot (\mathbf{B_2} - \mathbf{B_1}) = 0 \\ \hat{n} \times (\mathbf{E_2} - \mathbf{E_1}) = 0 \\ \hat{n} \times (\mathbf{H_2} - \mathbf{H_1}) = \mathbf{k_f} \end{cases} \tag{3}$$

$\sigma_f = \dfrac{dQ_f}{ds}$, free charge surface density.
$k_f = \dfrac{dI_f}{dl}$, free current linear density.
$\hat{n}$ points from medium 1 to medium 2.

1.2 d'Alembert Equation

Relation of potentials and electromagnetic field:

$$\begin{cases} \mathbf{E} = -\nabla \phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \mathbf{B} = \nabla \times \mathbf{A} \end{cases} \tag{4}$$

d'Alembert Equation

$$\Box A^\mu = -\mu_0 J^\mu \tag{5}$$

$$\begin{cases} \Box = -\dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2} + \nabla^2 = \partial_\mu \partial^\mu \\ A^\mu = (\phi/c,\ \mathbf{A}) \\ J^\mu = (\rho c,\ \mathbf{J}) \\ \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{cases}$$

$$\begin{cases} \partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t},\ \nabla \right) \\ \partial^\mu = \left(-\frac{1}{c}\frac{\partial}{\partial t},\ \nabla \right) \end{cases}$$

Continuity Equation:

$$\partial_\mu J^\mu = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \tag{6}$$

Lorenz gauge:

$$\partial_\mu A^\mu = \frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{A} = 0 \tag{7}$$

Retarded Potential

$$\begin{cases} t_{\text{ret}} = t - |\mathbf{r} - \mathbf{r}'|/c \\ A^\mu(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \frac{J^\mu(\mathbf{r}', t_{\text{ret}})}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \end{cases} \tag{8}$$

1.3 Covariant Maxwell Equations

Electromagnetic field tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{9}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix} \tag{10}$$

$F^{\mu\nu}: F^{0i} = F_{i0},\ F^{i0} = F_{0i},\ F^{ij} = F_{ij}$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \tag{11}$$

Covariant Maxwell Equations

$$\begin{cases} \partial_\mu F^{\mu\nu} = \Box A^\nu = -\mu_0 J^\nu \\ \partial_{[\mu} F_{\nu\rho]} = \partial_\mu F_{\nu\rho} + \partial_\rho F_{\mu\nu} + \partial_\nu F_{\rho\mu} = 0 \end{cases} \tag{12}$$

2 Dielectric and Magnetic Media

2.1 Electric Polarization

Molecular electric dipole moment:

$$\mathbf{p} = q\mathbf{l} \tag{13}$$

Electric polarization intensity vector:

$$\mathbf{P} = \frac{\sum_i \mathbf{p}_i}{\Delta V} = \frac{N_{\text{total}}\, q\mathbf{l}}{\Delta V} = Nq\mathbf{l} \tag{14}$$

Bound charge: charge distribution arising from polarization

$$\begin{aligned} Q_p &= -\oint Nq\mathbf{l}\cdot d\mathbf{S} = -\oint \mathbf{P}\cdot d\mathbf{S} \\ &= -\int \nabla\cdot\mathbf{P}\, dV \end{aligned}$$

Polarized volume bound charge density:

$$\rho_p = \frac{dQ_p}{dV} = -\nabla\cdot\mathbf{P} \tag{15}$$

Polarized surface bound charge surface density: between medium 2 and medium 1, $\hat{n}$: from $1 \to 2$

$$\sigma_p = \frac{dQ_{p2} - dQ_{p1}}{dS} = -\frac{(\mathbf{P}_2 - \mathbf{P}_1)\cdot d\mathbf{S}}{dS} = -\hat{n}\cdot(\mathbf{P}_2 - \mathbf{P}_1) \tag{16}$$

Free charge density:

$$\begin{aligned} \rho_f &= \varepsilon_0 \nabla\cdot\mathbf{E} - \rho_p \\ &= \varepsilon_0 \nabla\cdot\mathbf{E} + \nabla\cdot\mathbf{P} \\ &= \nabla\cdot(\varepsilon_0\mathbf{E} + \mathbf{P}) = \nabla\cdot\mathbf{D} \end{aligned}$$

2.2 Magnetization

i: small loop current; $\mathbf{a}$: loop area vector.

Magnetic moment:

$$\mathbf{m} = i\mathbf{a} \tag{17}$$

Magnetization intensity:

$$\mathbf{M} = \frac{\sum_i \mathbf{m}_i}{\Delta V} = \frac{N_{\text{total}}\, i\mathbf{a}}{\Delta V} = Ni\mathbf{a} \tag{18}$$

Total magnetization current:

$$\begin{aligned} I_m &= \oint Ni\mathbf{a}\cdot d\mathbf{l} = \oint \mathbf{M}\cdot d\mathbf{l} \\ &= \int (\nabla\times\mathbf{M})\cdot d\mathbf{S} \end{aligned}$$

Magnetization current density:

$$\mathbf{J}_m = \nabla \times \mathbf{M} \tag{19}$$

Magnetization surface current density:

$$\alpha_m = \hat{n} \times (\mathbf{M}_2 - \mathbf{M}_1) \tag{20}$$

Polarization current density:

$$\mathbf{J}_p = \frac{\sum_i q_i \mathbf{v}_i}{\Delta V} = \frac{\partial \mathbf{P}}{\partial t} \tag{21}$$

Free current density:

$$\begin{aligned}
\mathbf{J}_f &= \frac{1}{\mu_0}\nabla \times \mathbf{B} - \mathbf{J}_m - \mathbf{J}_p - \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\
&= \frac{1}{\mu_0}\nabla \times \mathbf{B} - \nabla \times \mathbf{M} - \frac{\partial \mathbf{P}}{\partial t} - \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\
&= \nabla \times \left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M} \right) - \frac{\partial}{\partial t}\left(\varepsilon_0 \mathbf{E} + \mathbf{P}\right) \\
&= \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}
\end{aligned}$$

3 Action Principle

3.1 Action of a Charged Particle in EM Field

Action

$$S = \int L dt = \int \left(-mc^2 d\tau + qA_\mu dx^\mu\right) = \int \left(-\frac{mc^2}{\gamma} + qA_\mu \dot{x}^\mu\right) dt, \quad (\dot{x}^\mu = (c, \mathbf{v})) \tag{22}$$

Lagrangian

$$\begin{aligned} L &= -\frac{mc^2}{\gamma} + qA_\mu \dot{x}^\mu \\ &= -mc^2\sqrt{1 - \frac{v^2}{c^2}} + q\mathbf{v}\cdot\mathbf{A} - q\phi \end{aligned} \tag{23}$$

Euler-Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v_i}\right) = \frac{\partial L}{\partial x_i} \tag{24}$$

$$\iff \begin{cases} \dfrac{d\mathbf{p}}{dt} = (q\mathbf{E} + q\mathbf{v}\times\mathbf{B}) \\ \mathbf{p} = \dfrac{m\mathbf{v}}{\sqrt{1-\frac{v^2}{c^2}}} = \gamma m\mathbf{v} \\ \gamma = \dfrac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \dfrac{dt}{d\tau} \end{cases} \tag{25}$$

Canonical Momentum:

$$\mathbf{P} = \frac{\partial L}{\partial \mathbf{v}} = \mathbf{p} + q\mathbf{A} \tag{26}$$

Hamiltonian

$$\begin{aligned} H(\mathbf{x}, \mathbf{P}) &= \mathbf{P}\cdot\mathbf{v} - L \\ &= \gamma mc^2 + q\phi \\ &= \sqrt{(\mathbf{P} - q\mathbf{A})^2 c^2 + m^2c^4} + q\phi \end{aligned} \tag{27}$$

Hamilton canonical equations

$$\begin{cases} \mathbf{v} = \dfrac{\partial H}{\partial \mathbf{P}} \\ \dot{\mathbf{P}} = -\dfrac{\partial H}{\partial \mathbf{x}} = -\nabla H \end{cases} \tag{28}$$

$$\iff \frac{d\mathbf{p}}{dt} = q\mathbf{E} + q\mathbf{v}\times\mathbf{B} \tag{29}$$

3.2 Action of the Electromagnetic Field

Lagrangian density

$$\begin{aligned} \mathcal{L} &= -\frac{1}{4\mu_0} F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu \\ &= \frac{1}{2}\left(\varepsilon_0 E^2 - \frac{1}{\mu_0}B^2\right) - \rho\phi + \mathbf{J}\cdot\mathbf{A}. \end{aligned} \tag{30}$$

Euler-Lagrange equation

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0 \tag{31}$$

$$\iff \begin{cases} \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu \\ \partial_{[\rho} F_{\mu\nu]} = \partial_\rho F_{\mu\nu} + \partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} = 0. \end{cases} \tag{32}$$

Canonical Momentum Density

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} = -\frac{1}{\mu_0} F^{0\mu} = \left(0,\ -\frac{1}{\mu_0 c}\mathbf{E}\right) \tag{33}$$

Hamiltonian Density

$$\begin{aligned} \mathcal{H}(A_\mu,\ \Pi_\mu) &= \Pi^\mu(\partial_0 A_\mu) - \mathcal{L} \\ &= \Pi^i(\partial_0 A_i) - \mathcal{L} \end{aligned}$$

$$\begin{cases} \mathbf{E} = -\nabla\phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \nabla \cdot \mathbf{E} = \rho/\varepsilon_0 \end{cases}$$

$$\Rightarrow \mathcal{H} = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \mathbf{J}\cdot\mathbf{A} + \nabla\cdot(\varepsilon_0\phi\mathbf{E}) \quad \langle \text{Precise} \rangle \tag{34}$$

$$\because \int \nabla\cdot(\varepsilon_0\phi\mathbf{E}) dV = \oint \varepsilon_0\phi\mathbf{E}\cdot d\mathbf{S} \to 0,\ \text{omit.}$$

$$\therefore \mathcal{H} = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \mathbf{J}\cdot\mathbf{A} \tag{35}$$

$$\begin{aligned} \therefore d\mathcal{H} &= \varepsilon_0 \mathbf{E}\cdot d\mathbf{E} + \frac{1}{\mu_0}\mathbf{B}\cdot d\mathbf{B} - \mathbf{J}\cdot d\mathbf{A} + \varepsilon_0 \nabla\phi\cdot d\mathbf{E} \\ &= \varepsilon_0(\mathbf{E}+\nabla\phi)\cdot d\mathbf{E} - \mathbf{J}\cdot d\mathbf{A} + \frac{1}{\mu_0}\mathbf{B}\cdot d(\nabla\times\mathbf{A}) \\ &= -\frac{1}{c}(\mathbf{E}+\nabla\phi)\cdot d\ - \mathbf{J}\cdot d\mathbf{A} + \frac{1}{\mu_0}(\nabla\times\mathbf{B})\cdot d\mathbf{A} \\ &\quad - \frac{1}{\mu_0}\nabla\cdot(\mathbf{B}\cdot d\mathbf{A}) \quad \text{(omit)} \end{aligned}$$

Hamiltonian canonical equations

$$\begin{cases} \partial_0 A_i = \dfrac{\partial \mathcal{H}}{\partial \Pi^i} \\ \partial_0 \Pi^i = -\dfrac{\partial \mathcal{H}}{\partial A_i} \end{cases} \tag{36}$$

$$\Rightarrow \begin{cases} \mathbf{E} = -\nabla\phi - \dfrac{\partial \mathbf{A}}{\partial t} \\ \nabla\times\mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t} \end{cases} \tag{37}$$

$$(\mathbf{B} = \mu_0\mathbf{H},\ \mathbf{D} = \varepsilon_0\mathbf{E})$$

4 Lorentz Transformation of EM Field

In this section we study how the electromagnetic field tensor $F^{\mu\nu}$ transforms under a Lorentz boost. Consider an inertial frame S' moving with velocity v along the x-axis relative to S. The Lorentz transformation matrix is

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}. \tag{38}$$

The electromagnetic field tensor and its components are defined as

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}, \tag{39}$$

with the identifications

$$\frac{E_i}{c} = F^{0i}, \qquad B_i = \frac{1}{2}\varepsilon_{ijk}F^{jk}. \tag{40}$$

Under a Lorentz transformation, the tensor transforms as

$$F^{\mu\nu'} = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}F^{\alpha\beta}, \qquad x^{\mu'} = \Lambda^{\mu}{}_{\alpha}x^{\alpha}. \tag{41}$$

In matrix notation this reads

$$F' = \Lambda F \Lambda^T = \Lambda F \Lambda, \tag{42}$$

since $\Lambda^T = \Lambda$ for this boost. Carrying out the matrix multiplication gives the transformed tensor explicitly:

$$F' = \begin{pmatrix} 0 & \dfrac{E_x}{c} & \dfrac{\gamma(E_y - vB_z)}{c} & \dfrac{\gamma(E_z + vB_y)}{c} \\ -\dfrac{E_x}{c} & 0 & \gamma\left(B_z - \dfrac{v}{c^2}E_y\right) & -\gamma\left(B_y + \dfrac{v}{c^2}E_z\right) \\ -\dfrac{\gamma(E_y - vB_z)}{c} & -\gamma\left(B_z - \dfrac{v}{c^2}E_y\right) & 0 & B_x \\ -\dfrac{\gamma(E_z + vB_y)}{c} & \gamma\left(B_y + \dfrac{v}{c^2}E_z\right) & -B_x & 0 \end{pmatrix}. \tag{43}$$

Comparing the components of F' with the definition of $F^{\mu\nu}$, we obtain the transformation rules for the electric and magnetic fields:

$$\begin{cases} E'_x = E_x \\ E'_y = \gamma(E_y - vB_z) \\ E'_z = \gamma(E_z + vB_y) \end{cases} \qquad \begin{cases} B'_x = B_x \\ B'_y = \gamma\left(B_y + \dfrac{v}{c^2}E_z\right) \\ B'_z = \gamma\left(B_z - \dfrac{v}{c^2}E_y\right) \end{cases} \tag{44}$$

For a boost along the x-axis, these relations can also be written in the following compact matrix forms:

$$\begin{pmatrix} E'_x/c \\ B'_x \end{pmatrix} = \begin{pmatrix} E_x/c \\ B_x \end{pmatrix},$$

$$\begin{pmatrix} E'_y/c \\ B'_z \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E_y/c \\ B_z \end{pmatrix},$$

$$\begin{pmatrix} E'_z/c \\ B'_y \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E_z/c \\ B_y \end{pmatrix}.$$

These equations show that electric and magnetic fields mix under Lorentz transformations, reflecting the unified nature of the electromagnetic field.

5 Static Fields

$r \gg r'$ far field:

$$\frac{1}{|\mathbf{r}-\mathbf{r}'|} \approx \frac{1}{r} - \mathbf{r}' \cdot \nabla \frac{1}{r} + \frac{1}{2}(\mathbf{r}' \cdot \nabla)^2 \frac{1}{r} \tag{45}$$

Potential in static field:

$$\begin{cases} \phi(\mathbf{r}) = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r' \\ \mathbf{A}(\mathbf{r}) = \dfrac{\mu_0}{4\pi} \displaystyle\int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r' \end{cases} \tag{46}$$

5.1 Static Electric Field

Multipole moments:

Charge:

$$Q = \int \rho(\mathbf{r}')\, d^3r' \tag{47}$$

Electric dipole moment:

$$\mathbf{p} = \int \rho(\mathbf{r}')\, \mathbf{r}'\, d^3r' \tag{48}$$

Electric quadrupole moment:

$$D_{ij} = \int 3\, r_i' r_j'\, \rho(\mathbf{r}')\, d^3r' \tag{49}$$

Scalar Potential:

$$\begin{cases} \phi(\mathbf{r}) \approx \phi^{(0)}(\mathbf{r}) + \phi^{(1)}(\mathbf{r}) + \phi^{(2)}(\mathbf{r}) \\ \phi^{(0)}(\mathbf{r}) = \dfrac{Q}{4\pi\varepsilon_0 r} \\ \phi^{(1)}(\mathbf{r}) = \dfrac{\mathbf{p}\cdot\hat{r}}{4\pi\varepsilon_0 r^2} \\ \phi^{(2)}(\mathbf{r}) = \dfrac{1}{24\pi\varepsilon_0} \displaystyle\sum_{i,j=1}^{3} D_{ij} \frac{\partial^2}{\partial r_i \partial r_j}\left(\frac{1}{r}\right) \end{cases} \tag{50}$$

Electric Field of Dipole:

$$\begin{aligned} \mathbf{E}^{(1)}(\mathbf{r}) &= -\nabla\phi^{(1)}(\mathbf{r}) \\ &= \frac{1}{4\pi\varepsilon_0} \frac{3(\mathbf{p}\cdot\hat{r})\hat{r} - \mathbf{p}}{r^3} - \frac{1}{3\varepsilon_0}\, \mathbf{p}\, \delta^3(\mathbf{r}) \end{aligned} \tag{51}$$

5.2 Static Magnetic Field

Magnetic Dipole:

$$\mathbf{m} = I\mathbf{S} = \frac{1}{2} I \oint \mathbf{r}' \times d\mathbf{l} \tag{52}$$

Vector Potential:

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &\approx \mathbf{A}^{(0)}(\mathbf{r}) + \mathbf{A}^{(1)}(\mathbf{r}) \\ \mathbf{A}^{(0)}(\mathbf{r}) &= \frac{\mu_0 I}{4\pi r} \oint d\mathbf{l} = 0 \\ \mathbf{A}^{(1)}(\mathbf{r}) &= \frac{\mu_0}{4\pi r^2}(\mathbf{m} \times \hat{r}) \end{aligned} \tag{53}$$

Magnetic Field of Dipole:

$$\begin{aligned} \mathbf{B}^{(1)}(\mathbf{r}) &= \nabla \times \mathbf{A}^{(1)} \\ &= \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \hat{r})\hat{r} - \mathbf{m}}{r^3} + \frac{2}{3}\mu_0 \, \mathbf{m} \, \delta^3(\mathbf{r}) \end{aligned} \tag{54}$$

5.3 Ferromagnetism (铁磁性)

5.3.1 Magnetic Media (磁介质)

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H} = \mu_0\mu_r\mathbf{H} \tag{55}$$

$$\begin{cases} \chi_m < 0: \text{ diamagnet (抗磁体)} \\ 0 < \chi_m \ll 1: \text{ paramagnet (顺磁体)} \\ \chi_m \gg 1: \text{ ferromagnet (铁磁体).} \end{cases} \tag{56}$$

5.3.2 Ferromagnetism (磁畴)

Magnetic domain: A tiny microscopic region inside ferromagnetic materials. Within one single domain, all atomic magnetic moments are aligned uniformly in the same direction.

Under external **B**, domain walls move and domains reorient, leading to magnetization.

$T > T_c$ (Curie temperature), domains disappear, ferromagnet turns into paramagnet.

$$T_c = \begin{cases} 770°C, \text{ Fe} \\ 1131°C, \text{ Co} \\ 358°C, \text{ Ni.} \end{cases} \tag{57}$$

Key property

$$\begin{cases} \text{large } \chi_m \\ \text{Nonlinear response} \\ \text{hysteresis (磁滞)} \end{cases} \tag{58}$$

5.3.3 Hysteresis Loop (磁滞回线)

M_s: Saturation (饱和) Magnetization
M_r: Remanent (剩余) Magnetization
H_c: Coercive (矫顽) Field
$M = 0$ at $H = -H_c$

$W = \oint HdB$: Hysteresis Loss.

5.3.4 Magnetic Circuit

Ampère's Law : $\oint \mathbf{H} \cdot d\mathbf{l} = NI$ Magnetic Flux : $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$

$$\Rightarrow \begin{cases} \text{Magnetic circuit law : } \ NI = \Phi_B R \\ \text{Magnetic Reluctance : } \ R = \dfrac{L}{\mu A} \end{cases} \tag{59}$$

6 Electromagnetic Wave

6.1 Wave Equation and Basic Properties

In vacuum, Maxwell equations lead to the wave equations for the electric and magnetic fields:

$$\Rightarrow \begin{cases} \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\mathbf{E} = 0 \\ \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\mathbf{B} = 0 \end{cases} \qquad \Rightarrow \begin{cases} \Box\mathbf{E} = 0 \\ \Box\mathbf{B} = 0 \end{cases} \tag{60}$$

The plane-wave solutions are

$$\Rightarrow \begin{cases} \mathbf{E} = E_0 \cos\left(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta\right)\hat{n} \\ \mathbf{B} = \frac{1}{c}\left(\hat{k}\times\mathbf{E}\right) \end{cases} \tag{61}$$

Phase velocity:

$$v_p = \frac{\omega}{k}$$

Group velocity:

$$v_g = \frac{d\omega}{dk}$$

6.2 Telegrapher Equation

For a uniform transmission line, let R, L, G, and C be the resistance, inductance, conductance, and capacitance per unit length. The voltage $V(z,t)$ and current $I(z,t)$ satisfy the telegrapher equations:

$$\begin{cases} \frac{\partial V}{\partial z} = -L\frac{\partial I}{\partial t} - RI \\ \frac{\partial I}{\partial z} = -C\frac{\partial V}{\partial t} - GV \end{cases} \tag{62}$$

Eliminating I yields the wave equation for V:

$$\frac{\partial^2 V}{\partial z^2} = LC\frac{\partial^2 V}{\partial t^2} + (RC + GL)\frac{\partial V}{\partial t} + RG\,V.$$

In the lossless case ($R = G = 0$), this reduces to the ideal wave equation:

$$\frac{\partial^2 V}{\partial z^2} = LC\frac{\partial^2 V}{\partial t^2}, \tag{63}$$

with phase velocity $v_p = 1/\sqrt{LC}$, which is the transmission-line analogue of $v_p = 1/\sqrt{\mu\varepsilon}$ for plane EM waves.

For time-harmonic fields $V(z,t) = \mathrm{Re}\{V(z)e^{i\omega t}\}$, the telegrapher equation becomes

$$\frac{d^2 V}{dz^2} = (R + i\omega L)(G + i\omega C)\,V = \gamma^2 V, \tag{64}$$

where $\gamma = \alpha + i\beta$ is the complex propagation constant. The attenuation constant α and phase constant β are given by

$$\gamma = \sqrt{(R + i\omega L)(G + i\omega C)} \quad \Longrightarrow \quad \begin{cases} \alpha = \mathrm{Re}(\gamma) \\ \beta = \mathrm{Im}(\gamma) \end{cases} \tag{65}$$

The characteristic impedance is

$$Z_0 = \sqrt{\frac{R + i\omega L}{G + i\omega C}}. \tag{66}$$

This formulation is directly analogous to the propagation of EM waves in conducting media, where the conductivity introduces an effective complex permittivity, linking naturally to the energy dissipation term $-\mathbf{E} \cdot \mathbf{J}$ in the Poynting theorem that follows.

6.3 Energy and Energy Flux (Poynting Theorem)

Poynting theorem:

$$\frac{\partial w}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J} \tag{67}$$

with

$$\begin{cases} w = \dfrac{1}{2}\left(\varepsilon E^2 + \dfrac{1}{\mu} B^2\right) \\ \mathbf{S} = \dfrac{1}{\mu} \mathbf{E} \times \mathbf{B} \end{cases} \tag{68}$$

For non-magnetic media ($\mu = \mu_0$) with refractive index n, the time-averaged intensity is

$$I = \langle S \rangle = \frac{1}{2} n \varepsilon_0 c E_0^2 \tag{69}$$

Power and total energy:

$$P = \int \mathbf{S} \cdot d\mathbf{A}, \qquad E_{\text{total}} = \int w \, dV$$

6.4 Interface Phenomena: Refraction, Reflection

Fermat's Principle

$$0 = \delta \int_S dt = \frac{1}{c} \delta \int_S n \, ds \quad \Longrightarrow \quad \text{Snell's Law} \tag{70}$$

Snell's Law

$$\begin{cases} n_1 \sin\theta_i = n_2 \sin\theta_t \\ \theta_i = \theta_r \end{cases} \tag{71}$$

Fresnel Equations From boundary conditions with $\sigma_f = k_f = 0$. Define polarisation relative to the incident plane:

- S polarization: $\mathbf{E} \perp$ incident plane
- P polarization: $\mathbf{E} \parallel$ incident plane

S Polarization:

$$\begin{cases} r_s = \dfrac{E_r}{E_i} = -\dfrac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} \\ t_s = \dfrac{E_t}{E_i} = \dfrac{2\cos\theta_i \sin\theta_t}{\sin(\theta_i + \theta_t)} \end{cases} \tag{72}$$

P Polarization:

$$\begin{cases} r_p = \dfrac{E_r}{E_i} = \dfrac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \\ t_p = \dfrac{E_t}{E_i} = \dfrac{2\cos\theta_i \sin\theta_t}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)} \end{cases} \tag{73}$$

Reflectance and transmittance:

$$R = r^2, \qquad T = 1 - R$$

Normal Incidence At normal incidence ($\theta_i \to 0$), the S/P distinction vanishes. Using Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$, we obtain

$$\boxed{r = \frac{n_1 - n_2}{n_1 + n_2}}, \qquad \boxed{t = \frac{2n_1}{n_1 + n_2}}$$

Brewster Angle The Brewster angle is the incident angle for which $r_p = 0$. This occurs when the reflected and transmitted rays are orthogonal:

$$\theta_i + \theta_t = \frac{\pi}{2}.$$

Substituting into Snell's law gives

$$n_1 \sin\theta_i = n_2 \cos\theta_i \quad \Longrightarrow \quad \tan\theta_i = \frac{n_2}{n_1}.$$

Thus

$$\theta_B = \arctan\left(\frac{n_2}{n_1}\right).$$

6.5 Interference and Diffraction

Interference For two waves at a point P:

$$\begin{cases} u_P = u_A + u_B \\ u_A = u_{A0} e^{i(\omega_A t - k_A r_A)} \\ u_B = u_{B0} e^{i(\omega_B t - k_B r_B)} \end{cases} \tag{74}$$

Intensity is proportional to $\langle u u^* \rangle$.

If A and B are incoherent ($k_A \neq k_B,\ \omega_A \neq \omega_B$):

$$I_P = I_A + I_B.$$

If they are coherent ($k_A = k_B,\ \omega_A = \omega_B$):

$$I_P = I_A + I_B + 2\sqrt{I_A I_B}\,\cos(\phi_A - \phi_B).$$

Helmholtz Equation From the wave equation, assuming harmonic time dependence $\Psi(\mathbf{r},t) \propto e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$, we get

$$(\nabla^2 + k^2)\Psi = 0, \tag{75}$$

where Ψ stands for any component of $\mathbf{E}$ or $\mathbf{B}$.

The Green function satisfies

$$(\nabla^2 + k^2)G(\mathbf{R}) = -\delta(\mathbf{R}). \tag{76}$$

With spherical symmetry $\nabla^2 \to \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial}{\partial R}\right)$, the outgoing solution is

$$G(R) = \frac{e^{ikR}}{4\pi R}, \qquad \mathbf{R} = \mathbf{r}_s - \mathbf{r}_0. \tag{77}$$

Using Green's identity:

$$\begin{aligned}\Psi(\mathbf{r}_0) &= \int \Psi(\mathbf{r}_s)\,\delta(\mathbf{r}_s - \mathbf{r}_0)\,dV \\ &= \int \left(G\nabla^2\Psi - \Psi\nabla^2 G\right) dV.\end{aligned}$$

Fresnel–Kirchhoff Diffraction Formula

$$\Psi(\mathbf{r}_0) = \frac{1}{i\lambda}\oint \Psi(\mathbf{r})\frac{e^{ikr}}{r}\frac{\cos\theta_s + \cos\theta}{2}\,ds. \tag{78}$$

6.6 Polarization

For a monochromatic plane wave propagating along z:

$$\begin{cases} E_x = E_{x0}\cos(\omega t - kz + \phi_x) \\ E_y = E_{y0}\cos(\omega t - kz + \phi_y) \\ \delta = \phi_y - \phi_x \end{cases} \tag{79}$$

The general polarisation equation is

$$\left(\frac{E_x}{E_{x0}}\right)^2 + \left(\frac{E_y}{E_{y0}}\right)^2 - \frac{2E_xE_y}{E_{x0}E_{y0}}\cos\delta = \sin^2\delta. \tag{80}$$

Polarisation states:

$$\begin{cases} \text{Linear}: \ \delta = 0 \text{ or } \pi \\ \text{Circular}: \ \delta = \pm\dfrac{\pi}{2} \text{ and } E_{x0} = E_{y0} \\ \text{Elliptical}: \ \text{otherwise} \end{cases} \tag{81}$$

Stokes parameters:

$$\begin{cases} I = E_{x0}^2 + E_{y0}^2 \\ Q = E_{x0}^2 - E_{y0}^2 \\ U = 2E_{x0}E_{y0}\cos\delta \\ V = 2E_{x0}E_{y0}\sin\delta \end{cases} \tag{82}$$

Degree of polarisation:

$$P = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}. \tag{83}$$

7 Electromagnetic Radiation

Retarded Potentials

$$\begin{cases} \Box A^\mu = -\mu_0 J^\mu \\ A^\mu = \left(\dfrac{\phi}{c}, \mathbf{A}\right) \\ J^\mu = (\rho c, \mathbf{J}) \\ \Box = -\dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2} + \nabla^2 \end{cases} \tag{84}$$

$$\Rightarrow \begin{cases} t' = t - \dfrac{|\mathbf{r} - \mathbf{r}'|}{c} \\ A^\mu(\mathbf{r}, t) = \dfrac{\mu_0}{4\pi}\int \dfrac{J^\mu(\mathbf{r}', t')}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \end{cases} \tag{85}$$

7.1 Point Charge Radiation

For field : $r \gg r'$

Point particle :

$$\begin{cases} \rho = q \cdot \delta(\mathbf{r}') \\ \mathbf{J} = \rho \mathbf{v} = q\mathbf{v} \cdot \delta(\mathbf{r}') \end{cases} \tag{86}$$

7.1.1 Liénard–Wiechert Potentials

In rest frame : $\mathbf{v} = 0$

$$\begin{cases} \tilde{\phi}(t) = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{\tilde{r}} \\ \tilde{\mathbf{A}}(t) = 0 \end{cases} \tag{87}$$

In moving frame : $\mathbf{v} \neq 0$

Lorentz transformation :

$$\begin{pmatrix} \phi/c \\ A^1 \\ A^2 \\ A^3 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{\phi}/c \\ \tilde{A}^1 \\ \tilde{A}^2 \\ \tilde{A}^3 \end{pmatrix} \tag{88}$$

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

$$\begin{cases} \tilde{r} = c(\tilde{t} - \tilde{t}') \\ \tilde{t} = \gamma\left(t - \dfrac{\beta \cdot \mathbf{x}}{c}\right) \\ \tilde{t}' = \gamma\left(t' - \dfrac{\beta \cdot \mathbf{x}'}{c}\right) \\ \mathbf{r} = \mathbf{x} - \mathbf{x}' \end{cases}$$

$$\Rightarrow \begin{cases} \dfrac{\partial t'}{\partial t} = \dfrac{1}{1 - \dfrac{\mathbf{v}\cdot\hat{r}}{c}} \\ \nabla t' = -\dfrac{\hat{r}}{c\left(1 - \dfrac{\mathbf{v}\cdot\hat{r}}{c}\right)} \end{cases} \tag{89}$$

$$\Rightarrow \begin{cases} \phi = \dfrac{q}{4\pi\varepsilon_0\left(r - \dfrac{\mathbf{v}\cdot\mathbf{r}}{c}\right)} \\ \mathbf{A} = \dfrac{q\mathbf{v}}{4\pi\varepsilon_0 c^2\left(r - \dfrac{\mathbf{v}\cdot\mathbf{r}}{c}\right)} \end{cases} \quad \text{(Liénard–Wiechert Potential)} \tag{90}$$

7.1.2 Non-Relativistic Radiation Fields

In the non-relativistic limit $v \ll c$, the Liénard–Wiechert potentials reduce to the simpler forms

$$\begin{cases} \phi(\mathbf{r},t) \approx \dfrac{q}{4\pi\varepsilon_0 r}\left[1 + \dfrac{\mathbf{v}(t')\cdot\hat{r}}{c}\right] \\ \mathbf{A}(\mathbf{r},t) \approx \dfrac{q\mathbf{v}(t')}{4\pi\varepsilon_0 c^2 r} \end{cases} \tag{91}$$

where $r = |\mathbf{r} - \mathbf{r}'|$ and all quantities on the right are evaluated at the retarded time $t' = t - r/c$. Using $\nabla t' = -\hat{r}/c$ and $\partial t'/\partial t = 1$, we derive the electric field from

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}. \tag{92}$$

The gradient of the scalar potential gives the static Coulomb term:

$$\begin{aligned} -\nabla\phi &= -(\nabla\phi)_{t'} - \nabla t'\frac{\partial\phi}{\partial t'} \\ &= -\frac{q}{4\pi\varepsilon_0}\nabla\frac{1}{r} - \left(-\frac{\hat{r}}{c}\right)\cdot\frac{q}{4\pi\varepsilon_0 r}\cdot\frac{\mathbf{a}\cdot\hat{r}}{c} \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\,(\mathbf{a}\cdot\hat{r}) \end{aligned}$$

For the vector potential,

$$-\frac{\partial\mathbf{A}}{\partial t} = -\frac{q}{4\pi\varepsilon_0 c^2}\frac{d}{dt'}\left(\frac{\mathbf{v}}{r}\right) = -\frac{q}{4\pi\varepsilon_0 c^2}\left(\frac{\mathbf{a}}{r} + \mathbf{v}\frac{\partial}{\partial t}\frac{1}{r}\right),$$

with $\partial_t(1/r) = -(\hat{r}\cdot\mathbf{v})/r^2$. Keeping only the radiation term that falls off as $1/r$ and dropping the higher-order velocity corrections, we obtain

$$\begin{aligned} \mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\left[\hat{r}(\mathbf{a}\cdot\hat{r}) - \mathbf{a}\right] \\ &= \frac{q\hat{r}}{4\pi\varepsilon_0 r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\times(\hat{r}\times\mathbf{a}) \end{aligned} \tag{93}$$

so the full non-relativistic electric field is

$$\boxed{\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\frac{\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r}\times(\hat{r}\times\mathbf{a})}. \tag{94}$$

Take acceleration in z direction $\mathbf{a} = a\hat{z}$. We can rewrite as:

$$\begin{aligned}
\mathbf{E} &= \mathbf{E}_{\text{static}} + \mathbf{E}_{\text{rad}},\\
\mathbf{E}_{\text{static}} &= \frac{q}{4\pi\varepsilon_0 r^2}\hat{r},\\
\mathbf{E}_{\text{rad}} &= \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta}
\end{aligned} \tag{95}$$

For the magnetic field, we compute $\mathbf{B} = \nabla\times\mathbf{A}$. The gradient acts on $1/r$ and on the retarded velocity:

$$\nabla\times\left(\frac{\mathbf{v}}{r}\right) = \nabla\left(\frac{1}{r}\right)\times\mathbf{v} + \frac{1}{r}(\nabla\times\mathbf{v}).$$

The first term gives the non-relativistic magnetostatic field (velocity field):

$$\mathbf{B}_{\text{static}} = \frac{q}{4\pi\varepsilon_0 c^2}\frac{\mathbf{v}\times\hat{r}}{r^2}, \tag{96}$$

which decays as $1/r^2$ and corresponds to the field of a uniformly moving charge. The second term yields the radiation field (acceleration field). Since

$$\nabla\times\mathbf{v} = \nabla t'\times\dot{\mathbf{v}} = -\frac{\hat{r}}{c}\times\dot{\mathbf{v}},$$

we find the radiation magnetic field

$$\mathbf{B}_{\text{rad}} = \frac{q}{4\pi\varepsilon_0 c^3 r}\,\dot{\mathbf{v}}\times\hat{r}. \tag{97}$$

Using the vector identity, this is exactly

$$\mathbf{B}_{\text{rad}} = \frac{1}{c}\,\hat{r}\times\mathbf{E}_{\text{rad}}. \tag{98}$$

Thus the complete non-relativistic magnetic field is

$$\boxed{\begin{aligned}
\mathbf{B} &= \frac{q}{4\pi\varepsilon_0 c^2}\frac{\mathbf{v}\times\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^3 r}\,\dot{\mathbf{v}}\times\hat{r}\\
&= \frac{1}{c^2}\mathbf{v}\times\mathbf{E}_{\text{static}} + \frac{1}{c}\hat{r}\times\mathbf{E}_{\text{rad}}
\end{aligned}} \tag{99}$$

Take acceleration in z direction $\mathbf{a} = a\hat{z}$. We can rewrite as:

$$\begin{aligned}
\mathbf{B} &= \mathbf{B}_{\text{static}} + \mathbf{B}_{\text{rad}},\\
\mathbf{B}_{\text{static}} &= \frac{q}{4\pi\varepsilon_0 c^2 r^2}\,(\mathbf{v}\times\hat{r}),\\
\mathbf{B}_{\text{rad}} &= \frac{qa\sin\theta}{4\pi\varepsilon_0 c^3 r}\,\hat{\phi}
\end{aligned} \tag{100}$$

The far-zone Poynting vector is therefore

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{1}{\mu_0 c} |\mathbf{E}|^2 \, \hat{r}. \tag{101}$$

With $|\mathbf{E}_{\rm rad}| = \dfrac{q\dot{v}}{4\pi\varepsilon_0 c^2 r} \sin\theta$, where θ is the angle between $\hat{r}$ and $\dot{\mathbf{v}}$, we have

$$\mathbf{S} = \frac{q^2 \dot{v}^2}{16\pi^2 \varepsilon_0 c^3 r^2} \sin^2\theta \, \hat{r}. \tag{102}$$

Integrating over a sphere,

$$P = \int \mathbf{S} \cdot d\mathbf{A} = \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \, \sin\theta \, r^2 (\mathbf{S} \cdot \hat{r}) = \frac{q^2 \dot{v}^2}{6\pi\varepsilon_0 c^3}, \tag{103}$$

which is the classical Larmor formula:

$$\boxed{P = \frac{q^2 \dot{v}^2}{6\pi\varepsilon_0 c^3}}. \tag{104}$$

Radiation Reaction Force To obtain the radiation reaction force, consider the energy balance over one period of motion. The work done by the reaction force must equal the negative of the energy radiated:

$$\int_0^T \mathbf{F}_{\rm rad} \cdot \mathbf{v} \, dt = -\int_0^T P \, dt = -\frac{q^2}{6\pi\varepsilon_0 c^3} \int_0^T \dot{\mathbf{v}}^{\,2} \, dt.$$

We now perform integration by parts on the right-hand side. Using the identity

$$\dot{\mathbf{v}}^{\,2} = \dot{\mathbf{v}} \cdot \dot{\mathbf{v}} = \frac{d}{dt}(\dot{\mathbf{v}} \cdot \mathbf{v}) - \mathbf{v} \cdot \ddot{\mathbf{v}},$$

and assuming periodic motion (or that the boundary term vanishes), we have

$$\int_0^T \dot{\mathbf{v}}^{\,2} \, dt = -\int_0^T \mathbf{v} \cdot \ddot{\mathbf{v}} \, dt.$$

Substituting this back gives

$$\int_0^T \mathbf{F}_{\rm rad} \cdot \mathbf{v} \, dt = \frac{q^2}{6\pi\varepsilon_0 c^3} \int_0^T \mathbf{v} \cdot \ddot{\mathbf{v}} \, dt = \int_0^T \left(\frac{q^2}{6\pi\varepsilon_0 c^3} \, \ddot{\mathbf{v}} \right) \cdot \mathbf{v} \, dt.$$

Since this equality holds for arbitrary velocity variations, we can read off the non-relativistic radiation reaction force (the Abraham–Lorentz force) as

$$\boxed{\mathbf{F}_{\rm rad} = \frac{q^2}{6\pi\varepsilon_0 c^3} \, \ddot{\mathbf{v}} = \frac{q^2}{6\pi\varepsilon_0 c^3} \, \dot{\mathbf{a}}}. \tag{105}$$

7.1.3 Relativistic Radiation Fields: Poynting Vector and Power

For relativistic motion, define the Lorentz factor

$$\boxed{\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-v^2/c^2}}}, \qquad \beta = \frac{\mathbf{v}}{c}. \tag{106}$$

For a charge moving with constant velocity (uniform rectilinear motion, $\dot{\beta} = 0$), the electric and magnetic fields at the observation point $\mathbf{R}$ (measured from the instantaneous position of the charge) are given by the exact expressions

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0} \frac{1-\beta^2}{(1-\beta^2\sin^2\theta)^{3/2}} \frac{\mathbf{R}}{R^3}, \qquad \mathbf{B} = \frac{1}{c}\,\beta \times \mathbf{E}, \tag{107}$$

where θ is the angle between $\mathbf{R}$ and β, and $R = |\mathbf{R}|$. These fields contain no radiation (no $1/R$ far-zone energy flux) because the Poynting vector vanishes in the far zone.

When the charge accelerates ($\dot{\beta} \neq 0$), the radiation (acceleration) fields are obtained from the full Liénard–Wiechert fields by keeping only the terms that fall off as $1/r$:

$$\mathbf{E}_{\rm rad}(\mathbf{r},t) = \frac{q}{4\pi\varepsilon_0 c} \frac{\hat{r} \times [(\hat{r}-\beta)\times\dot{\beta}]}{r(1-\beta\cdot\hat{r})^3}, \qquad \mathbf{B}_{\rm rad} = \frac{1}{c}\hat{r} \times \mathbf{E}_{\rm rad}, \tag{108}$$

where $\dot{\beta} = d\beta/dt$, and all quantities are evaluated at the retarded time.

The far-zone Poynting vector is therefore

$$\mathbf{S} = \frac{1}{\mu_0 c}|\mathbf{E}_{\rm rad}|^2\,\hat{r} = \frac{q^2}{16\pi^2\varepsilon_0 c} \frac{|\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]|^2}{r^2(1-\beta\cdot\hat{r})^6}\,\hat{r}. \tag{109}$$

The angular distribution of the radiated power is

$$\frac{dP}{d\Omega} = r^2\mathbf{S}\cdot\hat{r} = \frac{q^2}{16\pi^2\varepsilon_0 c} \frac{|\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]|^2}{(1-\beta\cdot\hat{r})^6}. \tag{110}$$

Integrating over all solid angles yields the relativistic Larmor formula (Liénard's formula)

$$P = \int \frac{dP}{d\Omega}\,d\Omega = \frac{q^2}{6\pi\varepsilon_0 c}\,\gamma^6\left[\dot{\beta}^2 - (\beta\times\dot{\beta})^2\right], \tag{111}$$

or equivalently, in terms of acceleration $\mathbf{a} = \dot{\mathbf{v}}$,

$$\boxed{P = \frac{q^2}{6\pi\varepsilon_0 c^3}\,\gamma^6\left[a^2 - \frac{(\mathbf{v}\times\mathbf{a})^2}{c^2}\right]}. \tag{112}$$

In the low-velocity limit $v \ll c$, $\gamma \simeq 1$ and the term $(\mathbf{v}\times\mathbf{a})^2/c^2$ is negligible, so this reduces to the classical Larmor formula.

7.1.4 Complete Liénard–Wiechert Fields

The full electric and magnetic fields of a point charge in arbitrary relativistic motion are

$$\mathbf{E}(\mathbf{r},t) = \frac{q}{4\pi\varepsilon_0}\,\frac{1}{(1-\beta\cdot\hat{r})^3}\left[\frac{\hat{r}-\beta}{\gamma^2 r^2} + \frac{\hat{r}\times[(\hat{r}-\beta)\times\dot{\beta}]}{cr}\right], \tag{113}$$

$$\mathbf{B}(\mathbf{r},t) = \frac{1}{c}\,\hat{r} \times \mathbf{E}(\mathbf{r},t), \tag{114}$$

where again all quantities on the right-hand side are evaluated at the retarded time $t' = t - r/c$, with $r = |\mathbf{r} - \mathbf{r}'|$ and $\hat{r} = (\mathbf{r} - \mathbf{r}')/r$. The first term in the brackets is the velocity field (generalized Coulomb field), which decays as $1/r^2$; the second term is the acceleration field (radiation field), which decays as $1/r$ and dominates in the far zone. In the non-relativistic limit $\beta \ll 1$, the above expressions reduce to the familiar forms

$$\mathbf{E} \simeq \frac{q}{4\pi\varepsilon_0}\frac{\hat{r}}{r^2} + \frac{q}{4\pi\varepsilon_0 c^2 r}\,\hat{r} \times (\hat{r} \times \dot{\mathbf{v}}), \qquad \mathbf{B} \simeq \frac{1}{c}\hat{r} \times \mathbf{E}, \tag{115}$$

recovering the static Coulomb field plus the radiation field given in Sec. 3.

7.2 Electric Dipole Radiation

Electric Dipole is also called as Hertzian Dipole.

Charge motion Consider a charge q oscillating along the z-axis with amplitude l:

$$\begin{cases} z(t') = l\sin(\omega t'), \quad l = \text{const} \\ v(t') = \dfrac{dz}{dt'} = \omega l\cos(\omega t') \\ a(t') = \dfrac{dv(t')}{dt'} = -\omega^2 l\sin(\omega t') \end{cases} \tag{116}$$

Fields and power (Larmor formula) For an observation point at distance r and polar angle θ (with respect to the dipole axis), the transverse electric field is

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0}\frac{\hat{\theta}}{c^2 r} \tag{117}$$

and the magnetic field $B = E/c$:

$$\mathbf{B}_{\text{rad}} = \frac{q}{4\pi\varepsilon_0 c^3 r}\,\mathbf{a} \times \hat{r}. \tag{118}$$

The instantaneous radiated power is given by the Larmor formula:

$$P(t) = \frac{q^2 a^2(t')}{6\pi\varepsilon_0 c^3}, \qquad t' = t - \frac{r}{c}, \qquad k = \frac{\omega}{c}. \tag{119}$$

Substituting $a(t') = -\omega^2 l\sin(\omega t')$, we obtain

$$\begin{aligned} E_{\text{rad}} &= \frac{ql\omega^2\sin\theta}{4\pi\epsilon c^2 r}\sin(kr - \omega t), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{q^2 l^2\omega^4}{6\pi\varepsilon_0 c^3}\sin^2(kr - \omega t). \end{aligned}$$

The time-averaged power is

$$\langle P\rangle = \frac{q^2 l^2\omega^4}{12\pi\varepsilon_0 c^3}. \tag{120}$$

Dipole picture Define the dipole moment amplitude $p_0 = ql$. Then

$$\begin{aligned} E_{\text{rad}} &= \frac{p_0 \omega^2 \sin\theta}{4\pi\epsilon c^2 r} \sin\left(kr - \omega t\right), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{p_0^2 \omega^4}{6\pi\varepsilon_0 c^3} \sin^2(kr - \omega t). \end{aligned} \tag{121}$$

The time-averaged power is

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\varepsilon_0 c^3}. \tag{122}$$

Current picture The current is the time derivative of the charge $q(t') = q\sin(\omega t')$:

$$I(t') = \frac{d}{dt'}\left[q\sin(\omega t')\right] = q\omega\cos(\omega t') = q\omega\cos(kr - \omega t). \tag{123}$$

Define the current amplitude

$$I_0 = q\omega. \tag{124}$$

Then

$$\begin{aligned} E_{\text{rad}} &= \frac{I_0 l \omega \sin\theta}{4\pi\epsilon c^2 r} \sin\left(kr - \omega t\right), \\ B_{\text{rad}} &= E_{\text{rad}}/c \\ P(t) &= \frac{I_0^2 l^2 \omega^2}{6\pi\varepsilon_0 c^3} \sin^2(kr - \omega t). \end{aligned}$$

and the average power becomes

$$\langle P \rangle = \frac{I_0^2 l^2 \omega^2}{12\pi\varepsilon_0 c^3}. \tag{125}$$

Define the radiation resistance R_{rad} such that

$$P(t) = I_0^2 R_{\text{rad}} \sin^2(kr - \omega t), \tag{126}$$

$$\langle P \rangle = \frac{1}{2} I_0^2 R_{\text{rad}}. \tag{127}$$

Comparing with the expression above,

$$R_{\text{rad}} = \frac{l^2\omega^2}{6\pi\varepsilon_0 c^3} = \frac{l^2}{6\pi\varepsilon_0 c^3}\left(\frac{2\pi c}{\lambda}\right)^2 = \frac{2}{3}\pi\mu_0 c\left(\frac{l}{\lambda}\right)^2 \approx 80\pi^2\,\Omega\,\left(\frac{l}{\lambda}\right)^2. \tag{128}$$

8 Radiation in Matter

Group and phase velocity For EM wave:

Group velocity is often under c:

$$v_g = \frac{d\omega}{dk} \tag{129}$$

More precisely, $v_g < c$ is typical for ordinary transparent media; in anomalous dispersion v_g can exceed c (or become negative), while the signal/front velocity remains bounded by c.

Phase velocity can be larger than c:

$$v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \tag{130}$$

Refractive index:

$$n = \frac{c}{v_p} \tag{131}$$

For most media $\mu \approx \mu_0$, so,

$$\frac{\varepsilon}{\varepsilon_0} \approx \frac{c^2}{v_p^2} = \frac{c^2 k^2}{\omega^2} \tag{132}$$

Wave equation with current density If there is no net charge $\rho = 0$, but charge particles can move (**J** exist), from Maxwell Equations, we can get wave equation with current density.

$$-\frac{1}{v_p^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} + \nabla^2 \mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{133}$$

Electric Field of EM wave:

$$\mathbf{E}(\mathbf{x}, t) = \mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)} \tag{134}$$

Take the direction of electric field as z axis:

$$\mathbf{E}_0 = E_0 \hat{\mathbf{z}} \tag{135}$$

We get:

$$(\frac{\omega^2}{v_p^2} - k^2)\mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{136}$$

Dispersion relation depends on relation between **J** and **E**

Full dynamical equation of electron

$$m_e \ddot{\mathbf{x}} + m_e \gamma_c \dot{\mathbf{x}} + m_e \omega_0^2 \mathbf{x} = -e(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}) + \mathbf{F}_{\text{rad}} \tag{137}$$

Here,

$m_e \ddot{\mathbf{x}}$: accelaration term.

$m_e \gamma_c \dot{\mathbf{x}}$: collision term.

$m_e \omega_0^2 \mathbf{x}$: harmonic oscillation term for bound electron.

$-e\mathbf{E}$: electric force term.

$-e\dot{\mathbf{x}} \times \mathbf{B}$: Lorentz force from magnetic field.

$\mathbf{F}_{\text{rad}} = q^2 \dot{\mathbf{a}}/6\pi\varepsilon_0 c^3$:radiation force reaction term.

Classical electron radius

$$r_{\mathrm{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{138}$$

Incident intensity of EM wave

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{139}$$

Radiation of electron accelerating in z axis

$$\mathbf{E}_{\mathrm{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{140}$$

Time-averaged Poynting vector

$$\langle \mathbf{S} \rangle = \langle S \rangle \hat{\mathbf{r}} = \varepsilon_0 c \langle E_{\mathrm{rad}}^2 \rangle \hat{\mathbf{r}} \tag{141}$$

Time-averaged total scattered power

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{A} = \int r^2 \langle S \rangle d\Omega \tag{142}$$

Scattering cross section

$$\sigma(\omega) = \frac{\langle P \rangle}{I_0} \tag{143}$$

Differential scattering cross section

$$\frac{d\sigma}{d\Omega} = \frac{1}{I_0}\frac{d\langle P \rangle}{d\Omega} = \frac{r^2 \langle S \rangle}{I_0} \tag{144}$$

Mean free path

$$\lambda_{\mathrm{mfp}} = \frac{1}{N\sigma} \tag{145}$$

N is the number density of electrons.

8.1 Wave in Plasma: Free Electron

Now we consider ionospheric (电离层) reflection

Wave equation with current density in vacuum becomes:

$$(\frac{\omega^2}{c^2} - k^2)\mathbf{E} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \tag{146}$$

Define N as electron number density. Current Density:

$$\mathbf{J} = -Ne\mathbf{v} \tag{147}$$

Electrons are influenced by EM field of EM wave, omit force due to radiation from electron acceleration. Non-relativistic case, dynamical equation of free electron:

$$m_e\ddot{\mathbf{x}} + m_e\gamma_c\dot{\mathbf{x}} = -e\mathbf{E} \tag{148}$$

Then,

$$\mathbf{x} = \frac{e\mathbf{E}}{m(\omega^2 + i\omega\gamma_c)}$$

$$\ddot{\mathbf{x}} = -\omega^2\mathbf{x} = -\frac{e\mathbf{E}}{m}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}$$

8.1.1 Dispersion relation of plasma

:

Def:

$$\omega_p^2 = \frac{Ne^2}{m_e\varepsilon_0} \tag{149}$$

Dispersion relation:

$$c^2k^2 = \omega^2 - \frac{\omega^2}{\omega^2 + i\omega\gamma_c}\omega_p^2 \tag{150}$$

If $\omega \gg \gamma_c$, dispersion relation is simplified:

$$c^2k^2 = \omega^2 - \omega_p^2 \tag{151}$$

When the EM wave travels from vacuum($n_1 = 1$) to refractive index n media:

$$n^2 = (\frac{c}{v_p})^2 = \frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2} \tag{152}$$

Propagation conditions in plasma

$$\begin{cases} \omega > \omega_p: & \text{EM wave propagate in Plasma.} \\ \omega < \omega_p: & \text{EM wave total reflection.} \end{cases} \tag{153}$$

8.1.2 Free-electron scattering

We now examine the radiation emitted by a single free electron driven by an incident EM wave, including the same damping term γ_c as in the plasma dispersion relation.

Acceleration of free electron due to EM wave:

$$\ddot{\mathbf{x}} = -\omega^2\mathbf{x} = -\frac{e\mathbf{E}}{m}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}$$

Radiation of accelerating electron:

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{154}$$

Def: the classical electron radius

$$r_{\text{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{155}$$

We get the far-zone radiation electric field :

$$\mathbf{E}_{\rm rad}(r,t) = E_0 \cos(kr - \omega t)\frac{r_{\rm e}}{r}\frac{\omega^2}{\omega^2 + i\omega\gamma_c}\sin\theta\,\hat{\boldsymbol{\theta}} \tag{156}$$

Incident intensity

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{157}$$

The time-averaged Poynting vector is:

$$\langle \mathbf{S} \rangle = \varepsilon_0 c \langle E_{\rm rad}^2 \rangle \hat{\mathbf{r}} = I_0 \frac{r_{\rm e}^2}{r^2}\frac{\omega^2}{\omega^2 + \gamma_c^2}\sin^2\theta\,\hat{\mathbf{r}} \tag{158}$$

and integrating over a sphere gives the total scattered power

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{A} = \frac{8\pi}{3} I_0 \frac{r_{\rm e}^2 \omega^2}{\omega^2 + \gamma_c^2} \tag{159}$$

Thus the scattering cross section is

$$\boxed{\sigma(\omega) = \frac{\langle P \rangle}{I_0} = \frac{8\pi}{3}\frac{r_{\rm e}^2\omega^2}{\omega^2 + \gamma_c^2}} \tag{160}$$

The differential scattering cross section is,

$$\frac{d\sigma}{d\Omega} = \frac{r^2\langle S \rangle}{I_0} = \frac{\omega^2}{\omega^2 + \gamma_c^2} r_e^2 \sin^2\theta \tag{161}$$

The frequency-dependent mean free path is therefore

$$\lambda_{\rm mfp}(\omega) = \frac{1}{N\sigma(\omega)} = \frac{3(\omega^2 + \gamma_c^2)}{8\pi N r_{\rm e}^2 \omega^2}. \tag{162}$$

Limiting cases:

For $\omega \gg \gamma_c$ (Thomson limit), the cross section becomes constant and the mean free path reduces to

$$\frac{d\sigma}{d\Omega} \to r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_{\rm e}^2, \qquad \lambda_{\rm mfp} \to \frac{3}{8\pi N r_{\rm e}^2}. \tag{163}$$

For $\omega \ll \gamma_c$ (strong damping), scattering is suppressed and the medium becomes effectively transparent:

$$\frac{d\sigma}{d\Omega} \to 0, \qquad \sigma \to 0, \qquad \lambda_{\rm mfp} \to \infty. \tag{164}$$

8.2 Wave in Dielectrics（电介质）: Bound Electron

Now we consider dielectrics deflection

Wave equation with current density in vacuum becomes:

$$(\frac{\omega^2}{c^2} - k^2)\mathbf{E} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \tag{165}$$

Define N as electron number density. Current Density:

$$\mathbf{J} = -Ne\mathbf{v} \tag{166}$$

Electrons are influenced by EM field of EM wave, omit force due to radiation from electron acceleration. Non-relativistic case, dynamical equation of bound electron in harmonic oscillation:

$$m_e\ddot{\mathbf{x}} + m_e\gamma_c\dot{\mathbf{x}} + m_e\omega_0^2\mathbf{x} = -e\mathbf{E} \tag{167}$$

Then,

$$\mathbf{x} = \frac{e\mathbf{E}}{m(\omega^2 - \omega_0^2 + i\omega\gamma_c)}$$

8.2.1 Dispersion relation of dielectrics

:

Def:

$$\omega_p^2 = \frac{Ne^2}{m_e\varepsilon_0} \tag{168}$$

Dispersion relation:

$$c^2k^2 = \omega^2 - \frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}\omega_p^2 \tag{169}$$

When the EM wave travels from vacuum($n_1 = 1$) to refractive index n media:

$$n^2 = (\frac{c}{v_p})^2 = \frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c} \tag{170}$$

8.2.2 Bound-electron scattering

We now examine the radiation emitted by a single bound electron (treated as a damped harmonic oscillator) driven by an incident EM wave, including the same damping term γ_c and a natural frequency ω_0.

Acceleration of bound electron due to EM wave:

$$\ddot{\mathbf{x}} = -\frac{e\mathbf{E}}{m_e}\frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}$$

Radiation of accelerating electron:

$$\mathbf{E}_{\text{rad}} = \frac{qa\sin\theta}{4\pi\varepsilon_0 c^2 r}\hat{\theta} \tag{171}$$

Def: the classical electron radius

$$r_{\text{e}} = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \tag{172}$$

We get the far-zone radiation electric field :

$$\mathbf{E}_{\text{rad}}(r,t) = E_0\cos(kr - \omega t)\frac{r_{\text{e}}}{r}\frac{\omega^2}{\omega^2 - \omega_0^2 + i\omega\gamma_c}\sin\theta\,\hat{\boldsymbol{\theta}} \tag{173}$$

Incident intensity

$$I_0 = \frac{1}{2}\varepsilon_0 c E_0^2 \tag{174}$$

The time-averaged Poynting vector is:

$$\langle \mathbf{S} \rangle = I_0 \frac{r_\mathrm{e}^2}{r^2} \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \sin^2\theta \, \hat{\mathbf{r}} \tag{175}$$

and integrating over a sphere gives the total scattered power

$$\langle P \rangle = \frac{8\pi}{3} I_0 \frac{r_\mathrm{e}^2 \omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \tag{176}$$

Thus the scattering cross section is

$$\boxed{\sigma(\omega) = \frac{\langle P \rangle}{I_0} = \frac{8\pi}{3} \frac{r_\mathrm{e}^2 \omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2}}. \tag{177}$$

The differential scattering cross section is,

$$\frac{d\sigma}{d\Omega} = \frac{r^2 \langle S \rangle}{I_0} = \frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2} \, r_e^2 \sin^2\theta \tag{178}$$

The frequency-dependent mean free path is therefore

$$\lambda_\mathrm{mfp}(\omega) = \frac{1}{N\sigma(\omega)} = \frac{3}{8\pi N r_\mathrm{e}^2} \frac{(\omega^2 - \omega_0^2)^2 + \omega^2 \gamma_c^2}{\omega^4}. \tag{179}$$

Limiting cases:

For $\omega \gg \omega_0$ and $\omega \gg \gamma_c$ (Thomson limit), the cross section becomes constant and the mean free path reduces to

$$\frac{d\sigma}{d\Omega} \to r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2, \qquad \lambda_\mathrm{mfp} \to \frac{3}{8\pi N r_\mathrm{e}^2}. \tag{180}$$

For $\omega \to \omega_0$ (resonant scattering peak), the cross section reaches its maximum and the mean free path becomes

$$\frac{d\sigma}{d\Omega} \to \frac{\omega_0^2}{\gamma_c^2} \, r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2 \frac{\omega_0^2}{\gamma_c^2}, \qquad \lambda_\mathrm{mfp} \to \frac{3\gamma_c^2}{8\pi N r_\mathrm{e}^2 \omega_0^2}. \tag{181}$$

For $\omega \ll \omega_0$ (Rayleigh scattering regime), the cross section scales as ω^4 and the mean free path scales as ω^{-4}:

$$\frac{d\sigma}{d\Omega} \to \frac{\omega^4}{\omega_0^4} \, r_e^2 \sin^2\theta, \qquad \sigma \to \frac{8\pi}{3} r_\mathrm{e}^2 \frac{\omega^4}{\omega_0^4}, \qquad \lambda_\mathrm{mfp} \to \frac{3\omega_0^4}{8\pi N r_\mathrm{e}^2 \omega^4}. \tag{182}$$

8.3 Wave in Conductors

Wave equation with current density in media becomes:

$$(\frac{\omega^2}{v_p^2} - k^2)\mathbf{E} = \mu \frac{\partial \mathbf{J}}{\partial t} \tag{183}$$

In linear, homogeneous, isotropic media, current density:

$$\mathbf{J} = \sigma \mathbf{E} \tag{184}$$

Dispersion relation of conducting media

$$k^2 = \mu\omega^2\left(\varepsilon + i\frac{\sigma}{\omega}\right) \tag{185}$$

Good or Poor Conductor

$$\begin{cases} \sigma \gg \varepsilon\omega, \text{ Good conductor}, & k \approx (1+i)\sqrt{\dfrac{\mu\omega\sigma}{2}} \\ \sigma \ll \varepsilon\omega, \text{ Poor conductor}, & k \approx \sqrt{\mu\varepsilon}\cdot\omega + i\frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \end{cases} \tag{186}$$

General Skin Depth Distance where wave amplitude decays to $1/e$ of its initial value.

$$e^{ikx} \to e^{i\mathrm{Re}(k)\cdot x}\cdot e^{-\mathrm{Im}(k)\cdot x}$$

From $\exp\left[-\mathrm{Im}(k)\cdot\delta\right] = 1/e$, we define general skin depth,

$$\delta = \frac{1}{\mathrm{Im}(k)} = \begin{cases} \sqrt{\dfrac{2}{\mu\omega\sigma}}, & \text{good conductor} \\ \dfrac{2}{\sigma}\sqrt{\dfrac{\varepsilon}{\mu}}, & \text{poor conductor} \end{cases} \tag{187}$$

9 Kramers–Kronig Relation

9.1 Cauchy Integral Formula

When α outside contour C,

$$I = \oint_C (z-\alpha)^n dz = 0 \tag{188}$$

When α inside contour C,

$$z = \alpha + Re^{i\varphi}$$

$$\begin{aligned} I &= \oint_C (z-\alpha)^n \, dz \\ &= \oint_C R^n e^{in\varphi} \, d\big(\alpha + Re^{i\varphi}\big) \\ &= iR^{n+1} \oint_C e^{i(n+1)\varphi} \, d\varphi \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\varphi} \, d\varphi \end{aligned}$$

$$I = \begin{cases} 0, & n \neq -1 \\ 2\pi i, & n = -1 \end{cases} \tag{189}$$

Therefore:

$$\frac{1}{2\pi i} \oint_C (z-\alpha)^n \, dz = 0, \qquad n \neq -1 \tag{190}$$

$$\frac{1}{2\pi i} \oint_C \frac{dz}{z-\alpha} = \begin{cases} 0, & \alpha \text{ outside } C \\ 1, & \alpha \text{ inside } C \end{cases} \tag{191}$$

Bounding the integral on the small circular arc: Let C_ε be a small circle of radius ε centered at α, with $z = \alpha + \varepsilon e^{i\varphi}$.

$$\begin{aligned} \left| \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha} \, dz \right| &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \left| \oint_{C_\varepsilon} dz \right| \\ &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \left| \int_0^{2\pi} i\varepsilon \, e^{i\varphi} \, d\varphi \right| \\ &\leq \frac{\max_{z \in C_\varepsilon} |f(z)-f(\alpha)|}{\varepsilon} \cdot \varepsilon \cdot 2\pi \\ &= 2\pi \cdot \max_{z \in C_\varepsilon} |f(z)-f(\alpha)| \end{aligned}$$

As $\varepsilon \to 0$, $\max_{z \in C_\varepsilon} |f(z)-f(\alpha)| \to 0$
Thus,

$$\lim_{\varepsilon \to 0} \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha} \, dz = 0.$$

Since the integrand is analytic in the region between the original contour C and the small circle C_ε, Cauchy's theorem gives

$$\oint_C \frac{f(z)-f(\alpha)}{z-\alpha}\,dz = \oint_{C_\varepsilon} \frac{f(z)-f(\alpha)}{z-\alpha}\,dz = 0$$

Cauchy Integral Formula:

$$f(\alpha) = \frac{1}{2\pi i}\oint_C \frac{f(\alpha)}{z-\alpha}\,dz = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-\alpha}\,dz \tag{192}$$

9.2 Sokhotski–Plemelj Equation

$$\frac{1}{x-i\varepsilon} = \frac{x+i\varepsilon}{x^2+\varepsilon^2} = \frac{x}{x^2+\varepsilon^2} + i\,\frac{\varepsilon}{x^2+\varepsilon^2}$$

$$\int_{-\infty}^{+\infty} \frac{\varepsilon}{x^2+\varepsilon^2}\,dx = \int_{-\infty}^{+\infty} \frac{1}{\left(\frac{x}{\varepsilon}\right)^2+1}\,d\Big(\frac{x}{\varepsilon}\Big) = \arctan\Big(\frac{x}{\varepsilon}\Big)\Big|_{-\infty}^{+\infty} = \pi$$

As $\varepsilon \to 0^+$:

$$\frac{\varepsilon}{x^2+\varepsilon^2} = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases}$$

So,

$$\lim_{\varepsilon\to 0^+} \frac{\varepsilon}{x^2+\varepsilon^2} = \pi\,\delta(x) \tag{193}$$

Cauchy principal value integral:

$$\mathcal{P}\int_a^b \frac{f(x)}{x-x_0}dx = \lim_{\varepsilon\to 0}\left(\int_a^{x_0-\varepsilon} \frac{f(x)}{x-x_0}dx + \int_{x_0+\varepsilon}^b \frac{f(x)}{x-x_0}dx\right) \tag{194}$$

Sokhotski–Plemelj Equation:

$$\lim_{\varepsilon\to 0^+} \frac{1}{x-i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) + i\pi\,\delta(x) \tag{195}$$

where $\mathcal{P}$ denotes the Cauchy principal value, and $\delta(x)$ is the Dirac delta function.

9.3 Kramers–Kronig Relation

As α inside contour C,

$$z \to w', \qquad \alpha \to w + i\varepsilon$$

Electric susceptibility as a complex number, must satisfy Cauchy Integral Formula:

$$\chi(w) = \frac{1}{2\pi i}\oint_C \frac{\chi(w')}{w'-w-i\varepsilon}\,dw' \tag{196}$$

Apply Sokhotski–Plemelj identity:

$$\begin{aligned}\chi(w) &= \frac{1}{2\pi i}\int_{-\infty}^{+\infty} \chi(w')\left[\mathcal{P}\left(\frac{1}{w'-w}\right) + i\pi\,\delta(w'-w)\right]dw' \\ &= \frac{1}{2\pi i}\int_{-\infty}^{+\infty} \chi(w')\,\mathcal{P}\left(\frac{1}{w'-w}\right)dw' + \frac{1}{2}\chi(w)\end{aligned}$$

Rearrange terms:

$$\chi(w) = \frac{1}{\pi i}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi(w')}{w'-w}\,dw' \tag{197}$$

Decompose susceptibility into real and imaginary parts:

$$\chi(w) = \chi_1(w) + i\,\chi_2(w)$$

Causality / Hermiticity condition:

$$\chi(-w) = \chi^*(w)$$

This gives symmetry relations:

$$\begin{cases} \chi_1(-w) = \chi_1(w) \\ \chi_2(-w) = -\chi_2(w) \end{cases}$$

χ_1 is even; χ_2 is odd function of frequency.

Real part of susceptibility:

$$\begin{aligned} \chi_1(w) &= \frac{1}{\pi}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi_2(w')}{w'-w}\,dw' \\ &= \frac{1}{\pi}\,\mathcal{P}\left[\int_0^{\infty} \chi_2(w')\left(\frac{1}{w'-w} + \frac{1}{w'+w}\right)dw'\right] \\ &= \frac{2}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{w'\,\chi_2(w')}{w'^2-w^2}\,dw' \end{aligned}$$

Imaginary part of susceptibility:

$$\begin{aligned} \chi_2(w) &= -\frac{1}{\pi}\,\mathcal{P}\int_{-\infty}^{+\infty} \frac{\chi_1(w')}{w'-w}\,dw' \\ &= -\frac{1}{\pi}\,\mathcal{P}\left[\int_0^{\infty} \chi_1(w')\left(\frac{1}{w'-w} - \frac{1}{w'+w}\right)dw'\right] \\ &= -\frac{2w}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{\chi_1(w')}{w'^2-w^2}\,dw' \end{aligned}$$

Relation to complex permittivity:

$$\chi(w) = \frac{\varepsilon(w)}{\varepsilon_0} - 1$$

Substitute $\chi_1 = \mathrm{Re}(\varepsilon/\varepsilon_0) - 1$, $\chi_2 = \mathrm{Im}(\varepsilon/\varepsilon_0)$:

$$\mathrm{Re}\left(\frac{\varepsilon(w)}{\varepsilon_0}\right) = 1 + \frac{2}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{w'\,\mathrm{Im}\left(\dfrac{\varepsilon(w')}{\varepsilon_0}\right)}{w'^2-w^2}\,dw' \tag{198}$$

$$\mathrm{Im}\left(\frac{\varepsilon(w)}{\varepsilon_0}\right) = -\frac{2w}{\pi}\,\mathcal{P}\int_0^{\infty} \frac{\mathrm{Re}\left(\dfrac{\varepsilon(w')}{\varepsilon_0}\right) - 1}{w'^2-w^2}\,dw' \tag{199}$$

These are the standard Kramers–Kronig dispersion relations, which connect the real dispersive part and imaginary absorptive part of complex relative permittivity.

Let $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega)$ be the complex refractive index, where n is the ordinary refractive index and κ is the extinction coefficient. Since

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = \big(\tilde{n}(\omega)\big)^2 = (n + i\kappa)^2 = n^2 - \kappa^2 + i\,2n\kappa,$$

we have

$$\begin{cases} \mathrm{Re}\left(\dfrac{\varepsilon(\omega)}{\varepsilon_0}\right) = n^2(\omega) - \kappa^2(\omega), \text{ related to dispersion and extinction} \\ \mathrm{Im}\left(\dfrac{\varepsilon(\omega)}{\varepsilon_0}\right) = 2n(\omega)\kappa(\omega), \text{ related to EM wave absorption.} \end{cases} \tag{200}$$

In the lossless limit ($\kappa \to 0$), the real part reduces to $n^2(\omega)$.